

Roll No. ....

Total Pages : 3

**CMCA/M-22**

**24527**

**BIG DATA AND PATTERN RECOGNITION**

Paper–MCA-20-41

Time Allowed : 3 Hours]

[Maximum Marks : 75

**Note :** Attempt **five** questions in all, selecting **one** question from each Unit. Question No. **1** is compulsory. All questions carry equal marks.

**Compulsory Question**

1. Write short notes on the following :
  - (a) What are the characteristics of Big Data?
  - (b) Write a short note on ICT developments that accelerate the pace of big data adoption.
  - (c) What do you mean by Failure? Discuss some of the failed standards.
  - (d) Write a short note on Zookeeper.
  - (e) Differentiate between Qualitative and Quantitative analysis.
  - (f) What are the paradigms of pattern recognition?
  - (g) What do you mean by schema-less model?
  - (h) Write short note on Tabular Store. 15

**UNIT-I**

2. (a) What is Data Analytics? Discuss different types of data analytics. 8

- (b) What are the different steps involved in the lifecycle of Big Data Analytics? Discuss. 7
3. (a) Discuss Business Architecture and explain how can Big Data be tied to the layers of business architecture. 8
- (b) Differentiate between traditional business intelligence and big data business intelligence. 7

**UNIT-II**

4. (a) What is Data Governance? Explain key concepts in the Governance approach for big data? 8
- (b) Discuss stepwise approach for Big Data Analysis. 7
5. Discuss different utilities available in Hadoop that are responsible for file system management, job control and cluster management. Explain Map Reduce programming model in detail. 15

**UNIT-III**

6. What are Pattern Recognition System? Discuss different approaches that can be used for pattern recognition and applications of pattern recognition. 15
7. (a) Define feature extraction and reduction. Also discuss different data domains for pattern recognition. 8
- (b) How can pattern recognition be done using nearest neighbour classifier? Explain. 7

**UNIT-IV**

8. (a) Define NoSQL and discuss the challenges of the traditional RDBMS approach that are overcome by NoSQL. 8

- (b) What are the different storage types available in NoSQL?  
Explain. 7
- 9. (a) Differentiate between Key-value store and document  
store. 8
- (b) How can different NoSQL products be compared? Also  
discuss the misconceptions about NoSQL. 7